Making atoms solve a puzzle

Challenge 02 — encode a graph in atom positions, photograph the answer

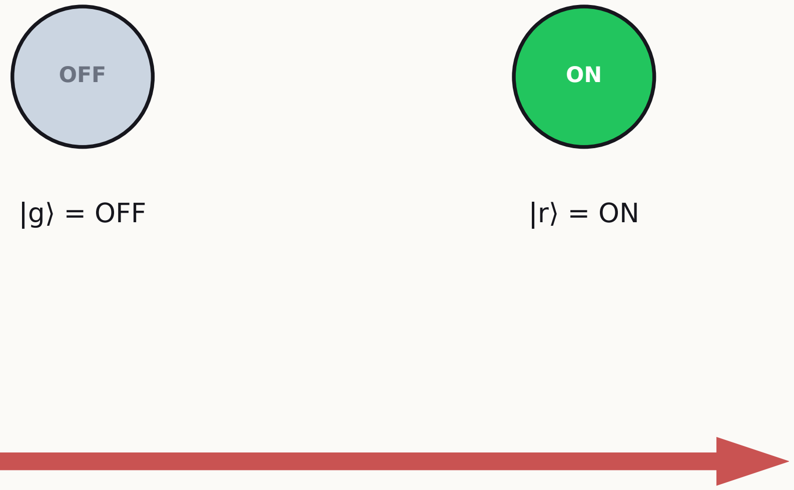
Success probability 0.727 / 0.657 (baseline) → 0.999998 / 0.999999 (ours)

Cloud: 1500 of 1500 photographs showed a correct answer · also run on real atoms

Team 6 · A Real Quantum Hackathon · Harmoniqs x Pasqal x Microsoft

THE BASICS

First: an atom is just a light switch



one laser shines on ALL atoms at once

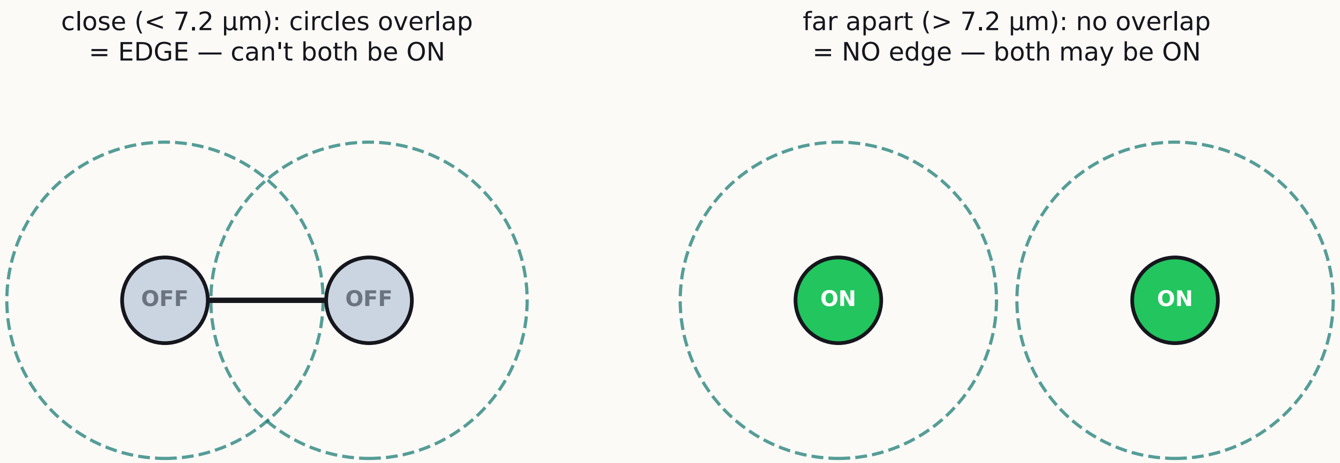
- Every atom is either OFF — the ground state $|g\rangle$ — or ON, the Rydberg state $|r\rangle$.
- We cannot poke atoms one by one: a single laser hits all of them together.
- We have just two knobs over time — the laser power $\Omega(t)$ and the frequency offset $\delta(t)$.
- Plus one more freedom: WHERE we place the atoms before anything starts.

The puzzle: seat the most people, no neighbors

- Classroom rule: no two students in adjacent seats. Question: what is the MOST students you can seat?
- That is the Maximum Independent Set (MIS) — a textbook 'hard' problem: the options explode as graphs grow.
- The trick of this challenge: atoms that sit close CANNOT both be ON (physics forbids it — the 'blockade').
- So the seating rule is enforced by nature, for free. Turn the laser sweep on, photograph the atoms: the ON atoms ARE the seated students.

We never check combinations one-by-one. The quantum system explores them together, and we shape the laser so the best answer is what the camera sees.

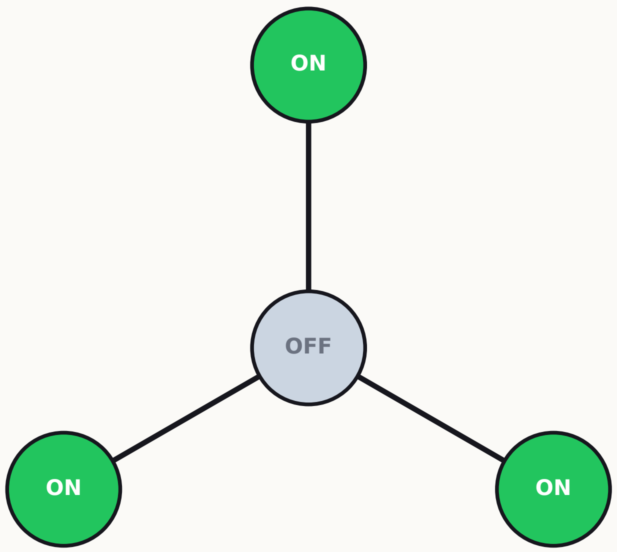
How atoms become a graph: distance decides the edges



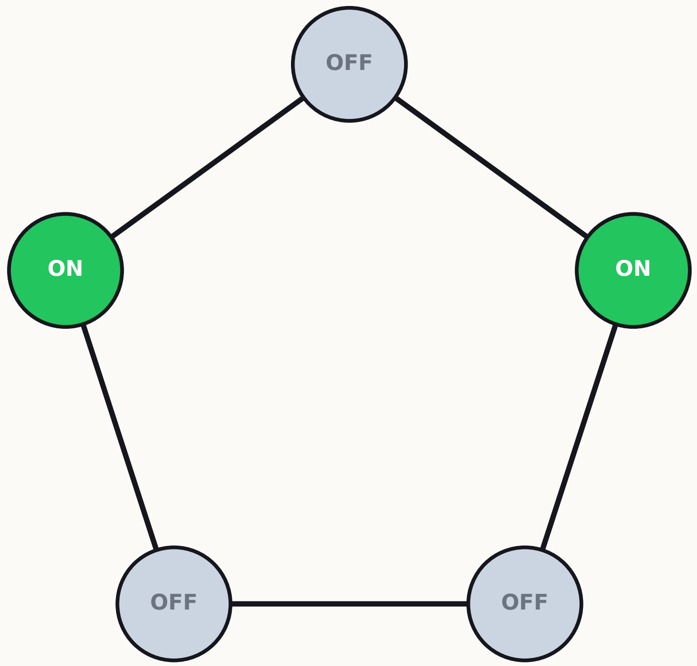
- Each atom becomes one dot — one vertex — of the graph.
- Any two atoms closer than the blockade radius R_b ≈ 7.2 µm form one edge.
- So we literally DRAW the puzzle with optical tweezers: the atom positions ARE the graph.
- And we verified every distance in code, so the register is exactly the target graph.

The two puzzles we were given

Score = P_MIS: the chance one photograph shows a best answer. Beat the starter sweep.



STAR: hub + 3 leaves.
Best answer: the 3 leaves ON, hub OFF (unique).



PENTAGON: 5 in a ring.
Best answer: any 2 non-neighbors ON — 5 equally good answers.

METHOD

The starter sweep vs what we did

- Starter sweep (the baseline): slowly tilt the laser frequency for 4000 ns — like carrying a full cup very carefully. Score: star 0.727, pentagon 0.657. It spills at the hardest moment.

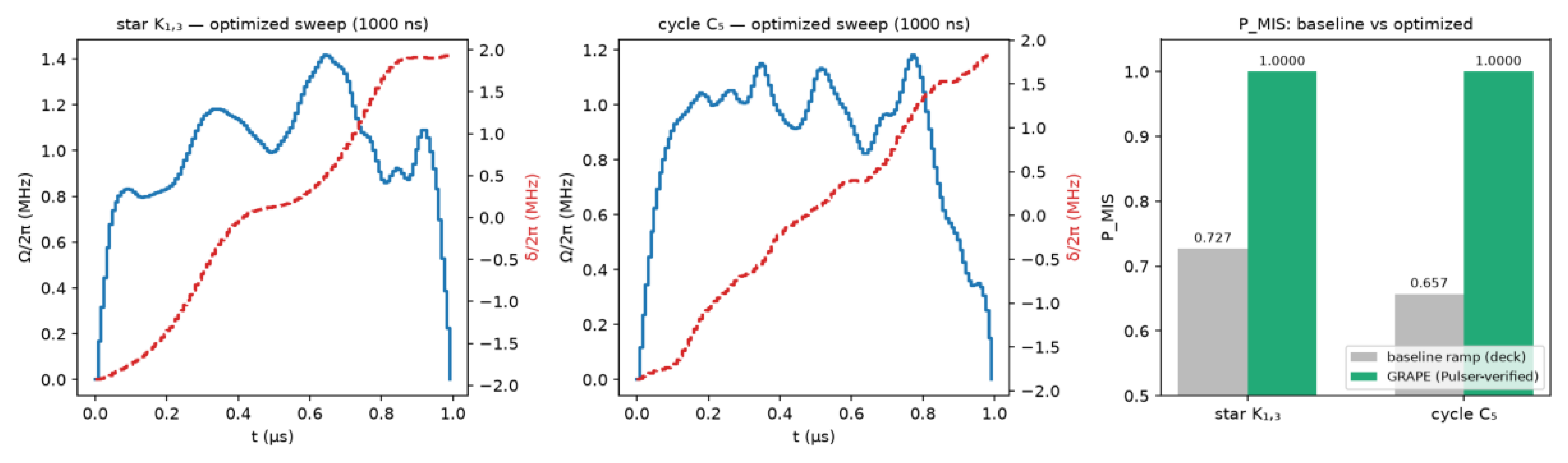
- Our approach: stop being careful, be OPTIMAL. We let a gradient optimizer (GRAPE) shape both laser knobs freely, to directly maximize the chance the photo shows a best answer.

- Key detail: the pentagon has FIVE equally-right answers. We optimize toward ALL of them at once (a 'projector' target) — the final state holds each answer at exactly 20%.

- Every result is double-checked in Pulser — the judges' own simulator (agreement: 7 decimal places).

Compute cost: all six optimizations ≈ 2 minutes on a laptop. Our sweeps are also 4× SHORTER than the baseline (1000 ns vs 4000 ns) — shorter = less time for real-world noise to corrupt the answer.

Result: near-perfect on both puzzles



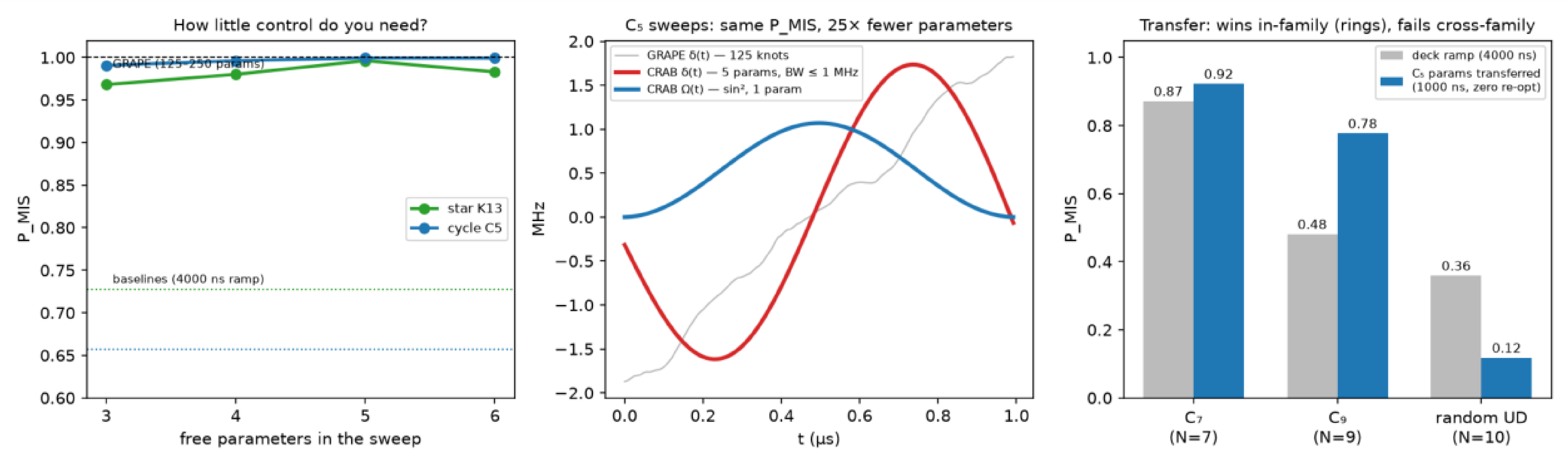
left/middle — x: time (µs), blue: laser power Ω, red: frequency offset δ · right — x: puzzle, y: success probability P_MIS

- The grey bars are the baseline, at 0.727 and 0.657; the green bars are ours, at 0.999998 and 0.999999.

Pasqal Cloud: 1500 photographs, 1500 correct answers

- We sent 3 optimized sweeps to Pasqal's cloud emulator, 500 shots each.
- Star: 500/500 photos showed exactly the right answer (0111 = hub OFF, leaves ON).
- For the pentagon at 1000 ns: again 500 out of 500 correct — split across ALL FIVE valid answers: 114, 98, 97, 96 and 95 counts. That is statistically even.
- That even split is a quantum signature: the machine doesn't pick one answer — it holds all five at once and samples them fairly.
- Pentagon (2000 ns): 500/500 again. Every batch ID is recorded in the repo for audit.

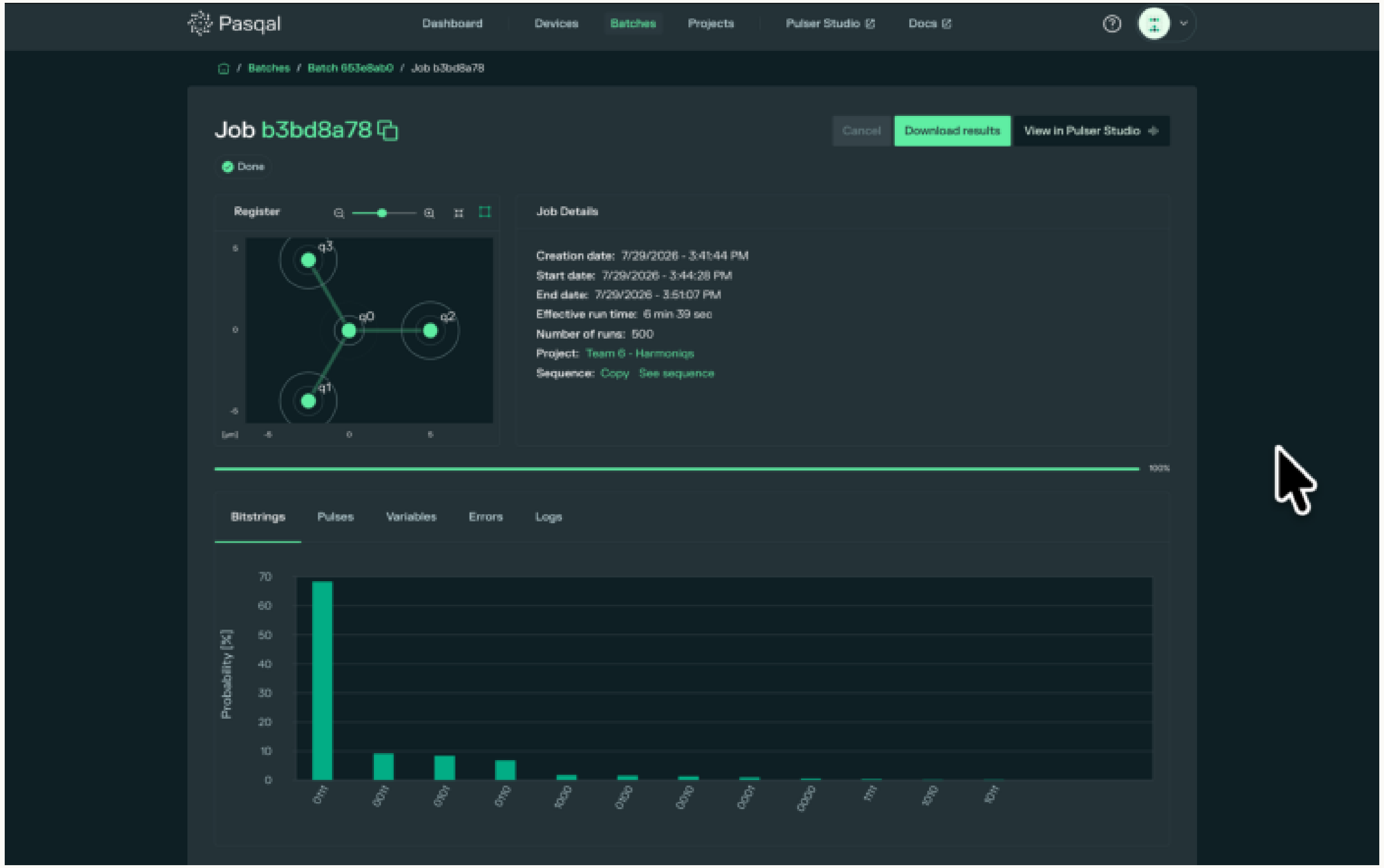
Then we asked: how simple can the winning sweep be?



left — x: number of knobs, y: P_MIS · middle — the 5-knob sweep vs the 125-knob one · right — x: new graphs, y: P_MIS with ZERO re-tuning

- Just FIVE knobs — instead of 125 — reach 0.996 to 0.999, found in about thirty seconds, and smooth enough for any hardware.
- The same 5 knobs, re-used unchanged on bigger rings: still beat the baseline (+0.30 on the 9-ring).
On a structurally different random graph they lose — honest limit, stated as a finding.

Real atoms, real answer: the machine drew our star

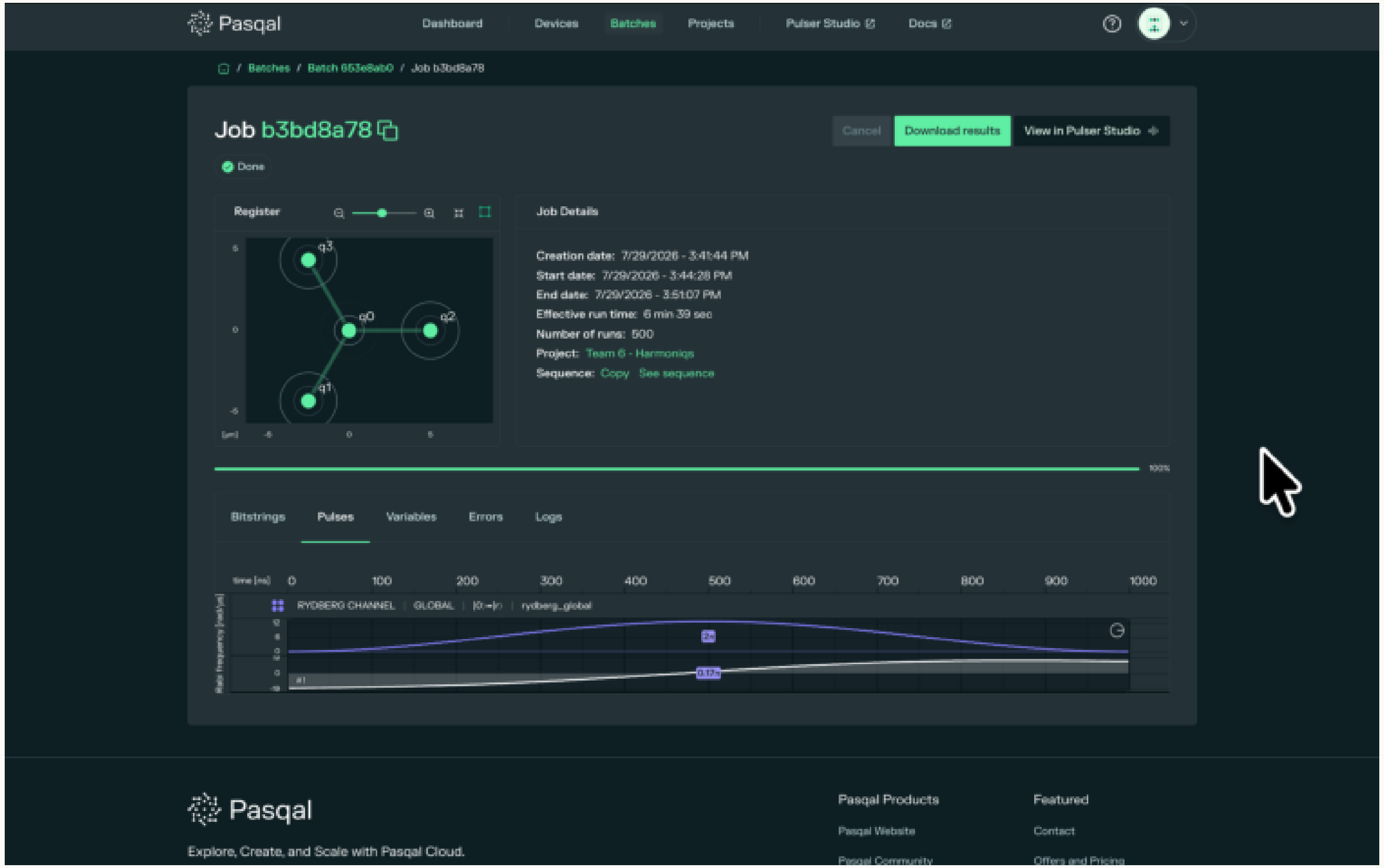


Pasqal's portal · top-left: OUR STAR, drawn by the machine's register viewer · bottom — x: measured pattern, y: % of 500 shots

- Look at the register panel: the hub q0 linked to q1 through q3 — Pasqal's own UI is showing the graph we encoded in atom positions.
- The towering bar is 0111 — the correct answer — at 68.4%, meaning 342 of 500 shots. Second place gets just 9.2%.
- So real atoms solve the puzzle decisively: the right answer wins by a factor of seven.
- This landed below our predicted 80–93% window: holding three fragile ON-atoms compounds readout loss. We published the prediction first, the analysis after.

REAL HARDWARE · THE PULSE

The sweep the machine actually played



Pasqal portal, Pulses tab · x: time 0–1000 ns · purple: laser power Ω(t) · white: frequency offset δ(t)

- This sweep is 1000 nanoseconds: one smooth bump of laser power, in purple — our five-knob, hardware-friendly design.
- The white curve tells the story: it starts NEGATIVE, so being ON is penalized, and ends POSITIVE, so being ON is rewarded — but only where the blockade allows it.
- That slow tilt walks the atoms from 'all OFF' to 'the best seating plan'.
- And its bandwidth is under 1 MHz by construction — well inside what the hardware can play.

The score, and why anyone should care

P_MIS = P(one photograph shows a best answer)

+0.34
our margin over baseline (pentagon)
the largest the scoring allows

1500/1500
cloud photographs showing
a correct answer

5 knobs
enough to reach 0.999 —
found in ~30 s on a laptop

- MIS is scheduling, spectrum allocation, portfolio screening — 'pick the most, no conflicts'. Rydberg machines solve it natively: the constraint is physics, not code.
- Our contribution scales: a 5-knob sweep, trained once on a small graph, re-used across a graph family at zero cost — the exact recipe Challenge 03's 80-atom instances need.
- Pasqal stack credit: Pulser device models + judge's simulator, free-tier cloud for 500-shot validation, and the FRESNEL_CAN1 QPU + portal for the real-atom evidence in this deck.

SUMMARY

Challenge 02 — the score sheet

	star $K_{1,3}$	pentagon C_5	duration
Baseline (starter sweep)	0.727	0.657	4000 ns
Ours · GRAPE (optimal control)	**0.999998**	**0.999999**	**1000–2000 ns**
Ours · 5-knob simple sweep	0.9962	0.9994	1000 ns
Cloud, 500 shots each	500/500	500/500 + 500/500	—
Real QPU, 500 shots	68.4% exact (7× runner-up)	n/a (lattice)	1000 ns

- Both puzzles beaten by +0.27 and +0.34 — the largest margins the scoring allows.
- Five equally-valid pentagon answers, sampled evenly — a visibly quantum result.
- Everything regenerates from the public repo: scorer, optimizer, figures, batch IDs.